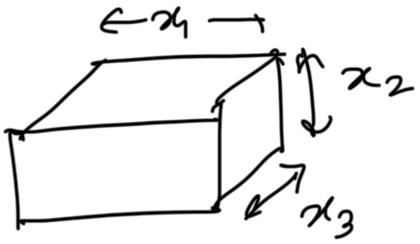
Constrained Optimization :

$$\min_{x} f(x)$$

subject to $g_i(x) \geq 0, \quad i = 1, 2, \ldots, m$ → inequality Constraints

and $h_j(x) = 0, \quad$ for $j = 1, 2, \ldots, p$ → equality Constraints

→ Design a rectangular box with minimum volume such that the total surface area is 8 sq. cm.



$$\min V = x_1 x_2 x_3$$

Subject to :

$$2(x_1 x_2 + x_2 x_3 + x_3 x_1) = 8$$

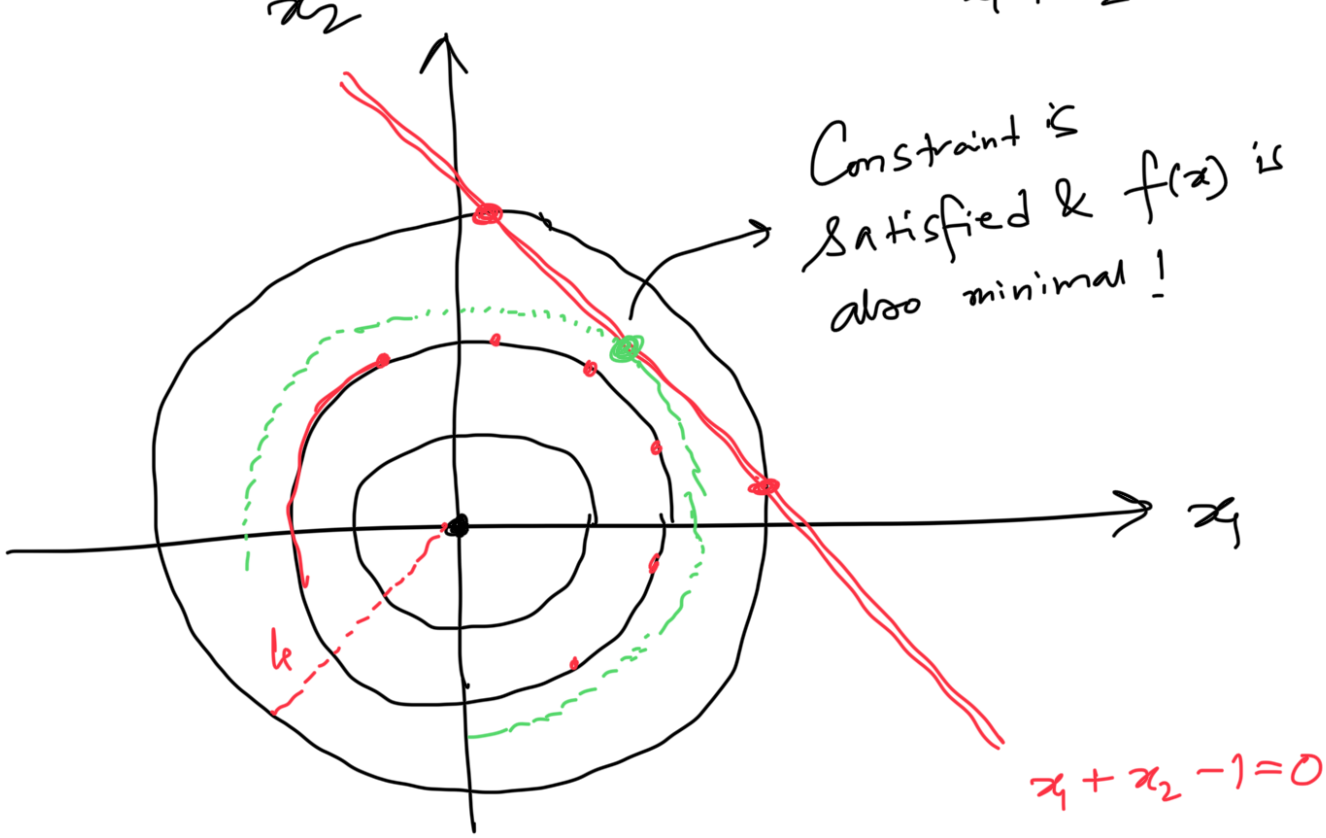
$$x_1 x_2 + x_2 x_3 + x_3 x_1 = 8/2 = 4$$

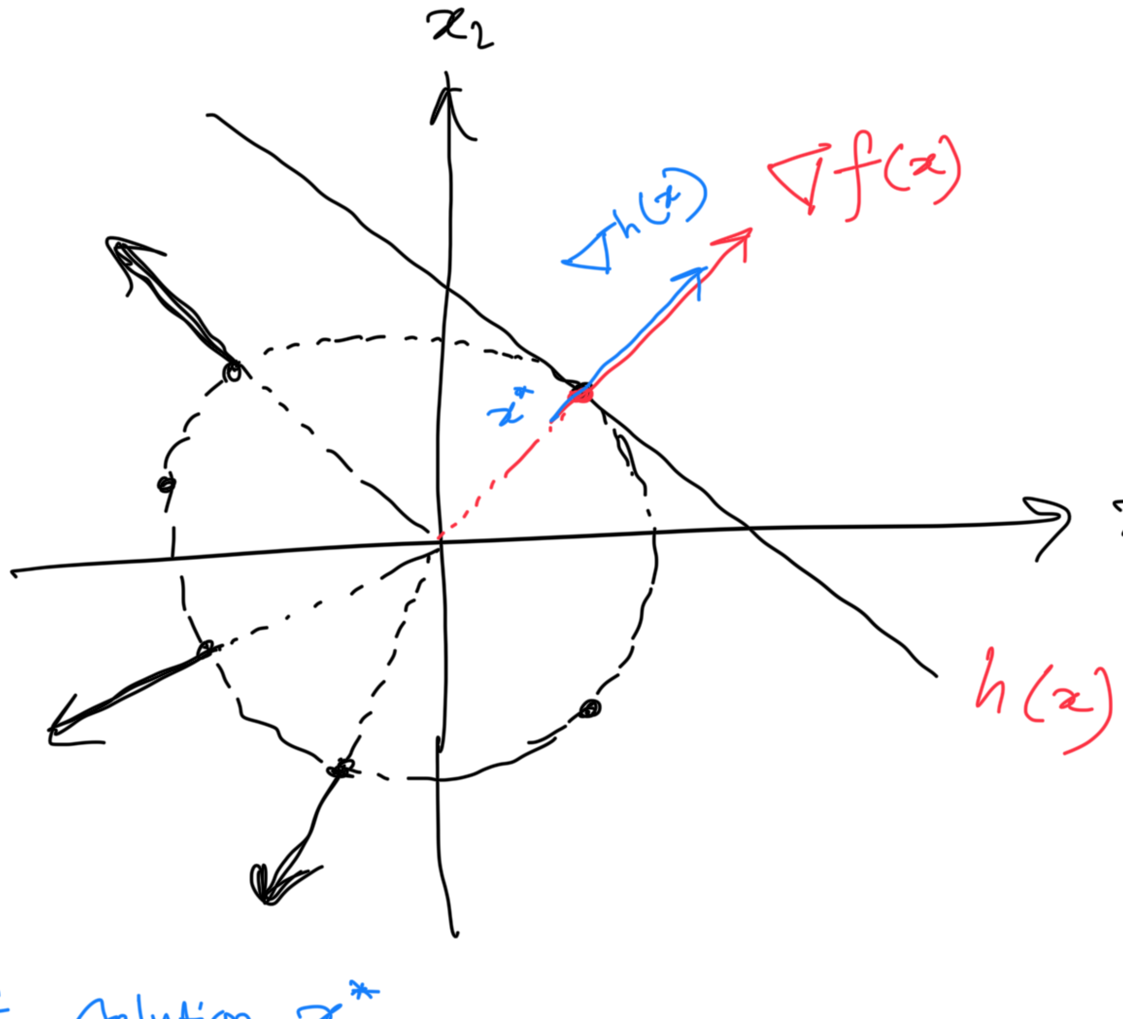
$$\min \; x_1^2 + x_2^2 = f(x_1, x_2) = f(x)$$

such that $x_1 + x_2 = 1$

$$\Rightarrow h(x) = x_1 + x_2 - 1 = 0$$

$x_1 + x_2 = k$





At solution x^*,

$\nabla f(x)$ & $\nabla h(x)$ share same direction !!

$$h(x) = w_1 x_1 + w_2 x_2 + b = 0$$

$$\nabla h(x) = \begin{bmatrix} \frac{\partial h(x)}{\partial x_1} \\ \frac{\partial h(x)}{\partial x_2} \end{bmatrix} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix} \rightarrow \perp^r \text{ to the St line } \underline{h(x) = 0}$$

i.e.,

$$\nabla f(x) = \underbrace{\lambda}_{\text{Scalar multiple (Lagrange's multiplier)}} . \nabla h(x)$$

$$\Rightarrow \nabla f(x) - \lambda . \nabla h(x) = 0$$

$$\Rightarrow \nabla \Big[\underbrace{f(x) - \lambda . h(x)} \Big] = 0$$

$$\Rightarrow \boxed{\nabla \mathcal{L}(x) = 0}$$

$$\mathcal{L}(x) = f(x) - \lambda . h(x) = \mathcal{L}(x, \lambda)$$

$\hookrightarrow$ Lagrangian function

$$f(x) = x_1^2 + x_2^2 \quad , \quad h(x) = x_1 + x_2 - 1$$

$$\mathcal{L}(x, \lambda) = (x_1^2 + x_2^2) - \lambda \cdot (x_1 + x_2 - 1)$$

$$\begin{cases} \dfrac{\partial \mathcal{L}}{\partial x_1} = 2x_1 - \lambda = 0 \Rightarrow x_1 = \lambda/2 \quad \text{--- } 1 \\[2ex] \dfrac{\partial \mathcal{L}}{\partial x_2} = 2x_2 - \lambda = 0 \Rightarrow x_2 = \lambda/2 \quad \text{--- } 2 \end{cases}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 \Rightarrow h(x) = 0 \Rightarrow x_1 + x_2 = 1$$

$$\Rightarrow \frac{\lambda}{2} + \frac{\lambda}{2} = 1$$

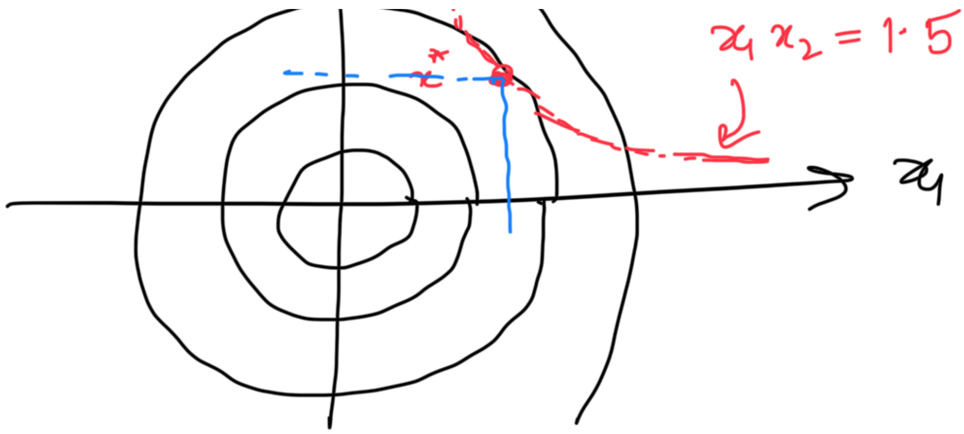
$$\Rightarrow \lambda = 1$$

$$\Rightarrow \boxed{x_1^* = 1/2 \ , \ x_2^* = 1/2}$$

Nonlinear equality

$$\min_x f(x) = x_1^2 + x_2^2$$

such that $2x_1 x_2 = 3 \Rightarrow x_2 = \dfrac{3}{2x_1}$



$$h(x) = x_1 x_2 - 1.5 = 0$$

$$\mathcal{L}(x) = x_1^2 + x_2^2 - \lambda (x_1 x_2 - 1.5)$$

$$\frac{\partial \mathcal{L}}{\partial x_1} = 2x_1 - \lambda x_2 = 0 \Rightarrow x_1 = \lambda x_2$$

$$\frac{\partial \mathcal{L}}{\partial x_2} = 2x_2 - \lambda x_1 = 0 \Rightarrow x_2 = \lambda x_1$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 \Rightarrow x_1 x_2 = 3/2$$

$$x_1 = \lambda x_2 = \lambda^2 x_1$$

$$\Rightarrow x_1 (1 - \lambda^2) = 0 \Rightarrow \lambda = 1 \quad (x_1, x_2 \neq 0)$$

$$\Rightarrow x_1^* = x_2^* = \sqrt{\frac{3}{2}}$$

Multiple equality & 1 inequality Constraint:

min $f(x)$ s.t. $h(x) = 0$

$$\min_x f(x) \quad \text{such that} \quad h_1(x) = 0$$
$$h_2(x) = 0$$
$$\vdots$$
$$h_m(x) = 0$$

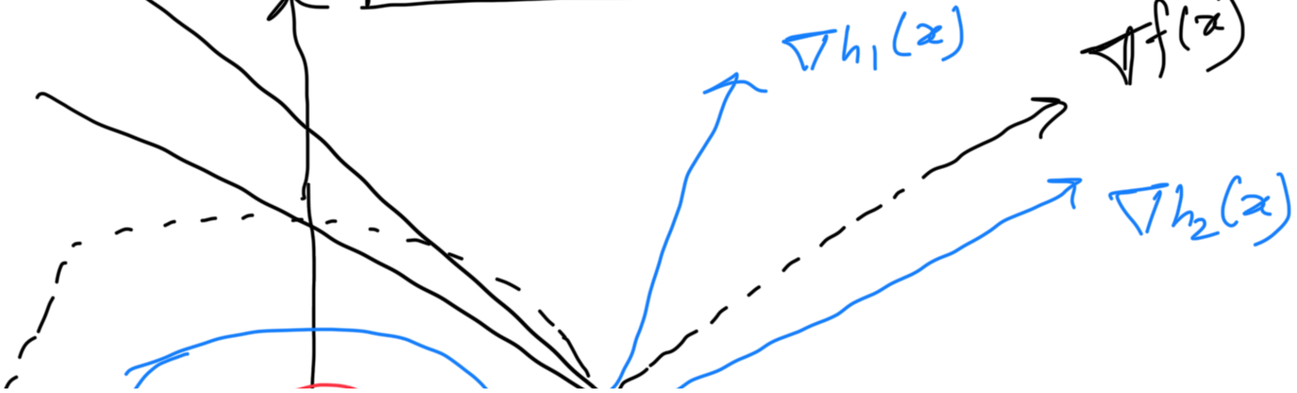
$$\text{and} \quad g(x) \geq b$$

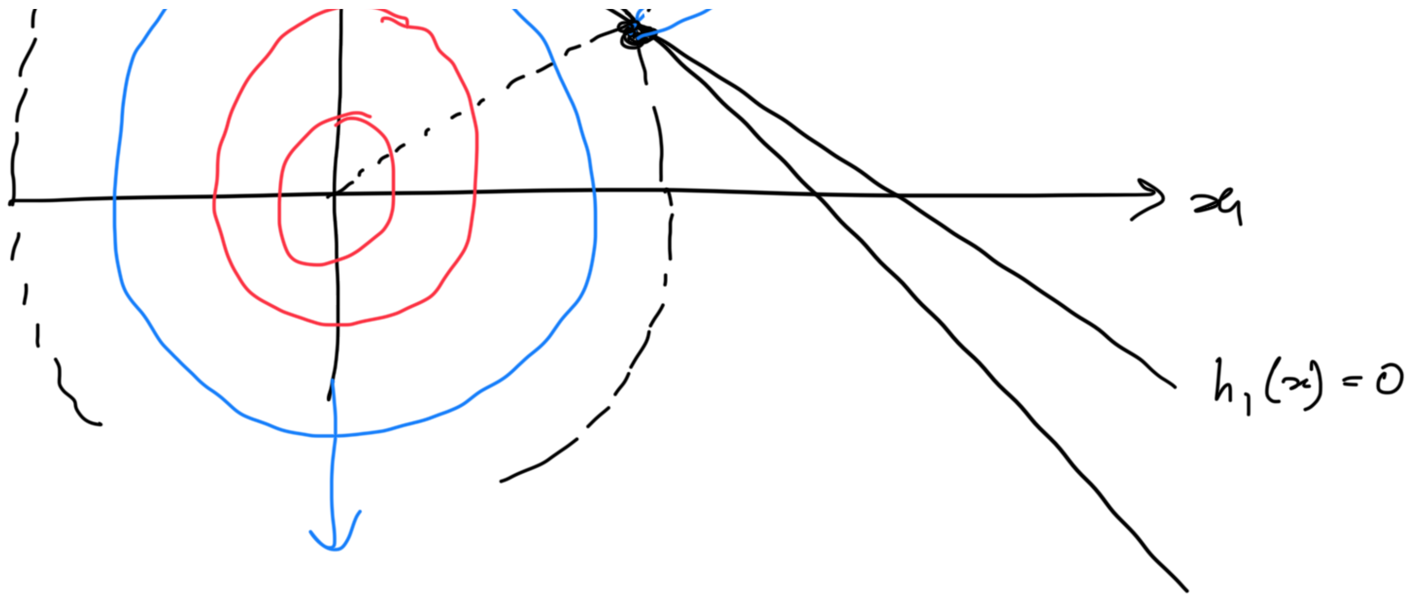
Sol$^{\underline{n}}$ approach: Ignore inequality constraint & solve the problem with multiple equality constraints.

If inequality constraint is satisfied at the sol$^{\underline{n}}$ point x^*, we are done!

If the inequality constraint is not satisfied, then the inequality constraint is BINDING & we need to enforce inequality constraint with equality i.e.

$$\boxed{g(x) = b}$$





$$\min_x \; x_1^2 + x_2^2 \quad \text{subject to} \quad h_1(x) = 3x_1 + 2x_2 - 6 = 0$$
$$h_2(x) = x_1 + x_2 - 1 = 0$$

at x^*

$$\nabla f(x) = \lambda_1 . \nabla h_1(x) + \lambda_2 \nabla h_2(x)$$

$$\Rightarrow \quad \nabla \Big[\underbrace{f(x) - \lambda_1 h_1(x) - \lambda_2 h_2(x)}_{\mathcal{L}(x, \lambda_1, \lambda_2)} \Big] = 0$$

$$\mathcal{L}(x, \lambda_1, \lambda_2) = f(x) - \lambda_1 h_1(x) - \lambda_2 h_2(x)$$

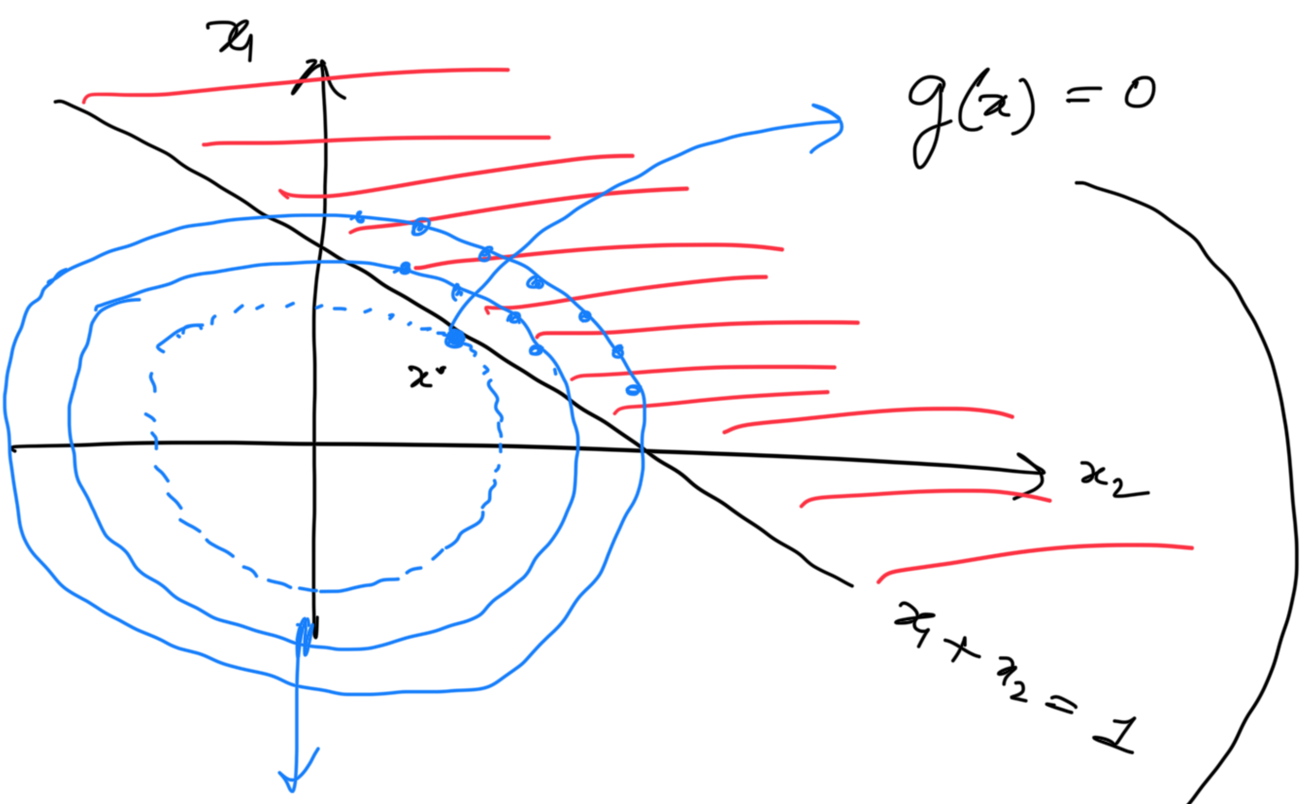
In general, for m inequality constraints,

$$\mathcal{L}(x, \bar{\lambda}) = f(x) - \sum_{i=1}^{m} \lambda_i h_i(x)$$

$$\underset{x}{\text{minimize}} \quad f(x) = x_1^2 + x_2^2$$

Such that $\quad x_1 + x_2 \geq 1$

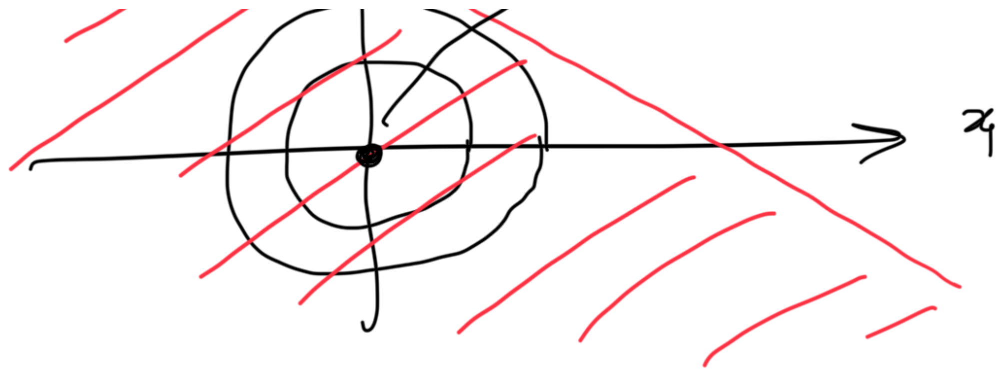
or $\quad g(x) = x_1 + x_2 - 1 \geq 0$



at x^* the inequality Constraint is Satisfied with equality. Provided the Constraint is binding !!

Our sol^n would be the un constrained minimum !!

$x^* = 0$

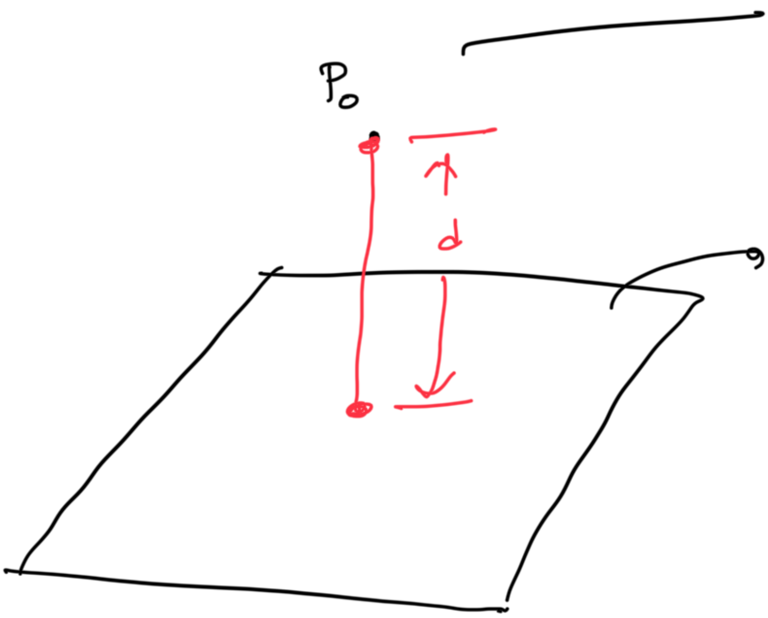


$$\min_{x} f(x) = x_1^2 + x_2^2 \quad \text{Such that} \quad x_1 + x_2 \leq 1$$

$$\text{or} \quad 1 \geq x_1 + x_2$$

$$\Rightarrow g(x) = 1 - x_1 - x_2 \geq 0$$

$\hookrightarrow$ The Constrain is not binding !!



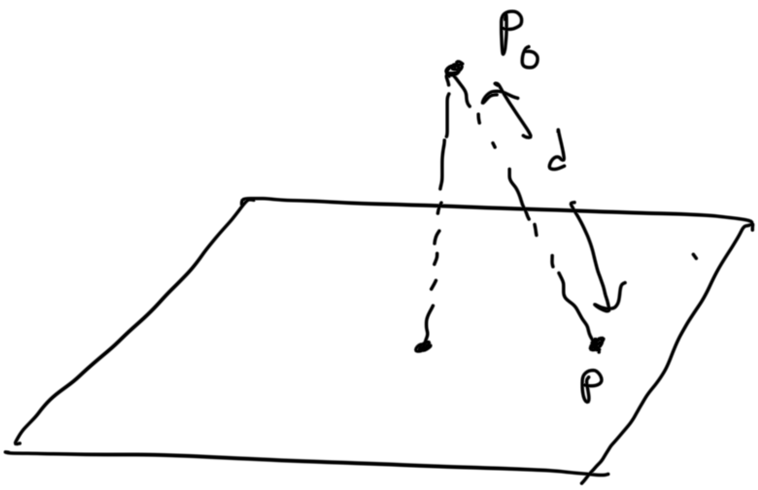
Uniform plane in 3D in $\mathbb{R}^3$

$$w_1 x_1 + w_2 x_2 + w_3 x_3 + b = 0 \Rightarrow \boxed{w^T x + b = 0}$$

$$P_0 = (x_1^0, x_2^0, x_3^0) \qquad \Rightarrow \underline{h(x) = 0}$$

What is the minimum Euclidean distance of

P_0 from the plane $w^T x + b = 0$



$P \rightarrow (x_1, x_2, x_3)$

$$d^2 = (x_1 - x_1^0)^2 + (x_2 - x_2^0)^2 + (x_3 - x_3^0)^2$$

$$\underset{x}{\text{minimize}} \quad d^2 \quad \text{such that} \quad w^T x + b = 0$$

$$\mathcal{L}(x_1, x_2, x_3, \lambda) = (x_1 - x_1^0)^2 + (x_2 - x_2^0)^2 + (x_3 - x_3^0)^2 - \lambda(w_1 x_1 + w_2 x_2 + w_3 x_3 + b)$$

$$\frac{\partial \mathcal{L}}{\partial x_1} = 2(x_1 - x_1^0) - \lambda w_1 = 0$$

$$\Rightarrow \quad x_1 - x_1^0 = \frac{\lambda w_1}{2}$$

$$x_2 - x_2^0 = \frac{\lambda w_2}{2}$$

$$x_3 - x_3^0 = \frac{\lambda w_3}{2}$$

$$\begin{cases} x_1 = x_1^0 - \dfrac{\lambda w_1}{2} \\ x_2 = x_2^0 - \dfrac{\lambda w_2}{2} \\ x_3 = x_3^0 - \dfrac{\lambda w_3}{2} \end{cases}$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 0 \Rightarrow w^T x + b = 0$$

$$\Rightarrow w_1 \left(x_1^0 - \frac{\lambda w_1}{2} \right) + w_2 \left(x_2^0 - \frac{\lambda w_2}{2} \right)$$

$$+ w_3 \left(x_3^0 - \frac{\lambda w_3}{2} \right) + b = 0$$

$$\Rightarrow \lambda = \frac{2 \left(w_1 x_1^0 + w_2 x_2^0 + w_3 x_3^0 + b \right)}{w_1^2 + w_2^2 + w_3^2}$$

$$d_{min} = \frac{\left| w_1 x_1^0 + w_2 x_2^0 + w_3 x_3^0 + b \right|}{\sqrt{w_1^2 + w_2^2 + w_3^2}}$$

$$d_{min} = \frac{|h(x)|}{\|w\|_2}$$